Software Engineering

BSc Computer Science - Durham University

Matthew Emerson

October 2026

Contents

1 Module Introduction **1**
1.1 Assessment 1
1.2 Timetable 1

2 Project Management **1**
2.1 Your Team, Formation, and Leadership 1
2.2 Team Engagement, Negotiation, and Ideation 2
2.3 Planning, Responsibilities, Milestones, and Risks 2

3 Software Engineering **3**

4 Legal, Social, Ethical, and Professional Issues **3**
4.1 Introduction to Software Engineering and Software Engineering Ethics 3

5 Group Project **4**

I Appendix **5**

6 Bibliography **5**
6.1 Module Professors 5
6.2 References 6
6.3 List of Figures 6
6.4 Reading List 6

1 Module Introduction

Software Engineering has the module code COMP2281 and is led by Dr Craig Stewart.

1.1 Assessment

This module is assessed through a group project, the deadlines for this can be seen in Table 1. Additional formatives are also shown in italics.

Title	Opening	Due	Feedback	Professor	Weight
Peer Evaluation 1	16th Oct	19th Nov	10th Dec	Dr Craig Stewart	1.67%
Requirement Documentation	16th Oct	19th Nov	17th Dec	Dr Craig Stewart	15%
Peer Evaluation 2	16th Oct	11th Feb	11th Mar	Dr Craig Stewart	1.67%
Product Presentation	Week 16	-	18th Mar	Dr Craig Stewart	15%
Technical Report	5th Feb	18th Mar	6th May	Dr Craig Stewart	30%
Peer Evaluation 3	5th Feb	18th Mar	6th May	Dr Craig Stewart	1.67%
Git Commit	5th Feb	29th Apr	1st Jul	Dr Craig Stewart	5%
Project Handover	5th Feb	29th Apr	1st Jul	Dr Craig Stewart	10%
Reflective Experience Report	5th Feb	6th May	1st Jul	Dr Craig Stewart	20%

Table 1: Assessments for Software Engineering.

1.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 2.

Day	Time	Class Type	Location	Lecturer
Tuesday	09:00 - 10:00	Lecture	TLC033	Dr Nelly Bencomo
Wednesday	11:00 - 12:00	Lecture	TLC033	Dr Nelly Bencomo
Thursday	11:00 - 13:00	Practical	CC-0007	Dr Gagangeet Aujla

Table 2: Michaelmas Term Timetable for Software Engineering.

These lectures are taught jointly by Dr Craig Stewart and Dr Nelly Bencomo.

2 Project Management

2.1 Your Team, Formation, and Leadership

This lecture was taught by Dr Nelly Bencomo.

Belbin (https://www.belbin.com/about/belbin-team-roles) defines nine roles someone can take in a team. These are:

- The Innovator who solves problems as the "ideas guy" of the group.
- The Resource Investigator who contacts external people/companies to bring additional resources to the group.
- The Coordinator who guides and controls the team. They know their team's abilities well.
- The Shaper who is a demanding and confrontational leader who pushes their members to excel.
- The Monitor/Evaluator who analyses proposed solutions and choices.
- The Team Worker who facilitates team functions and mediates issues.

LaTeX File Template v.2.3.0: Copyright 2026, Matthew Emerson

- The Implementor who implements the agreed upon plans.
- The Completer/Finisher who focuses on details and meeting deadlines.

After assigning roles and completing skill evaluation tests, the team can then create a CPD plan which identifies existing skills, areas for development, and a plan for doing that.

Maslow, A. (2016) splits competency into four categories: unconscious incompetence, conscious incompetence, unconscious competence, and finally conscious competence.

2.2 Team Engagement, Negotiation, and Ideation

This lecture was taught by Dr Nelly Bencomo.

When negotiating, you are more likely to be successful if you have the following factors: legitimacy, confidence, courtesy, adaptation to the other party, rapport, incentives/trade-offs, and research into the bigger picture.

2.3 Planning, Responsibilities, Milestones, and Risks

This lecture was taught by Dr Nelly Bencomo.

A simple task becomes a project when it produces a unique outcome, a finite timeline, interdependant work, and some level of uncertainty.

The project triangle states that there are always trade-offs between scope, resources, and time. This is because quality is never optional.

2.3.1 Planning

Plan outcomes before tasks:

- Goal: What useful result will the client recieve?
- Deliverable: What evidence or artefact demonstrates it?
- Milestone: When will we check progress?

Never adjust deadlines - its easier to instead negotiate scope.

RACI: A responsibilty assignment matrix for making decisions and producing work.

- Responsible: The person that does the work.
- Accountable: The person that owns the outcome/approval.
- Consulted: The person that provides input before action.
- Informed: The person who recieves progress updates.

Activity	Lead	Developer	Tester	Client
Agree Prject Scope	R	C	C	A
Implement a Feature	A	R	C	I
Test a Feature	A	C	R	I
Demonstrate Prototype	A	R	R	C

Table 3: RACI assignment matrix

LaTeX File Template v.2.3.0: Copyright 2026, Matthew Emerson

Project Management Tools: Trello (time management), GitHub Projects, Jira (scrum, sprints, backlogs), and/or Microsoft Planner (team tasks and deadlines).

Assigning priority level, using impact (1-5: negligible, minor, moderate, major, crucial) and risk (1-5: negligible, low, moderate, high, massive). Then set priority bands (e.g., 1-5 is low, 6-14 is medium, 15-25 is high).

5	5	10	15	20	25
4	4	8	12	16	20
3	3	6	9	12	15
2	2	4	6	8	10
1	1	2	3	4	5
Risk ↑ Impact →	**1**	**2**	**3**	**4**	**5**

Table 4: Assigning task priority in terms of risk and impact.

3 Software Engineering

4 Legal, Social, Ethical, and Professional Issues

This recommended reading for this sub-topic is:

- The Software Engineering Code of Ethics and Professional Practice, ACM (2024).

4.1 Introduction to Software Engineering and Software Engineering Ethics

This lecture was taught by Dr Craig Stewart.

4.1.1 Introduction to Software Engineering

Definition 4.1.1.1: An ill-structured problem, as defined by (Simon, 1973), is: elusive, unbounded, not understandable, without end conditions. They require orientation and context, and have many stakeholders and many possible solutions.

Definition 4.1.1.2: A good problem solver needs the following:

- **Method**: Refers to a formal procedure for accomplishing a goal, typically independent of the tools used.
- **Tool**: An instrument or automated system for accomplishing something in a better way.
- **Procedure**: A combination of tools and techniques to produce a product.
- **Paradigm**: The philosophy or approach for building a project.

Definition 4.1.1.3: For this module, a stakeholder will generally fall into one of three roles:

- **Customer/Client**: The company, organisation, or person who pays for the software system.
- **Developer**: The company, organisation, or person who is building the software system.
- **User**: The person/people who will actually use the system.

4.1.2 Introduction to Ethics

Definition 4.1.2.1: The software engineering code of ethics is defined by Rost and Glass (2010) and summarised more thoroughly by the Association for Computing Machinery (2024).

LaTeX File Template v.2.3.0: Copyright 2026, Matthew Emerson

5 Group Project

LaTeX File Template v.2.3.0: Copyright 2026, Matthew Emerson

Part I
Appendix

6 Bibliography

6.1 Module Professors

Module Lead: Doctor Craig Stewart

- Focus: Adaptive systems, HCI, and user modelling.
- Email: craig.d.stewart@durham.ac.uk
- Office Hours: Friday 11:00 - 12:00.

Lecturer: Doctor Nelly Bencomo

- Focus: Software engineering for autonomous, self-adaptive, AI and ML systems.
- Email: nelly.bencomo@durham.ac.uk
- Office Hours: Email to make an appointment.

Lecturer: Doctor Gagangeet Aujla

- Focus: Data-driven and data-intensive cyber-physical systems and digital twins.
- Email: gagangeet.s.aujla@durham.ac.uk
- Office Hours: Email to make an appointment.

Lecturer: Doctor Gokberk Kabacaoglu

- Focus: Machine learning and numerical applications.
- Email: gokberk.kabacaoglu@durham.ac.uk
- Office Hours: Thursday 14:00 - 15:00.

Lecturer: Doctor Ryan Crosby

- Focus: The assessment of novice programmers.
- Email: ryan.crosby@durham.ac.uk
- Office Hours: Email to make an appointment.

Lecturer: Doctor Sukanya Pandey

- Focus: Graph theory.
- Email: sukanya.pandey@durham.ac.uk
- Office Hours: Email to make an appointment.

6.2 References

6.2.1 Legal, Social, Ethical, and Professional Issues

Simon, H.A. (1973). The Structure of Ill Structured Problems. Artificial Intelligence, 4(3-4).

Rost, J. and Glass, R.L. (2010). The Dark Side of Software Engineering: Evil on Computing Projects. Washington, DC: IEEE Computer Society.

6.3 List of Figures

6.4 Reading List

6.4.1 Legal, Social, Ethical, and Professional Issues

Association for Computing Machinery (2024). The Software Engineering Code of Ethics and Professional Practice. [online] acm.org. Available at: https://www.acm.org/code-of-ethics/software-engineering-code [Accessed 7 Oct. 2026].